

## Lab 03 - LTspice

### Introduction

This will familiarize the student with the capabilities and basic use of LTSpice which is a freely available circuit simulation and analysis program, similar to many other Simulation Program with Integrated Circuit Emphasis (SPICE) programs (e.g. pSpice, ngSpice, etc).

### Part I: Introduction and Theory

LTSpice can be downloaded from:

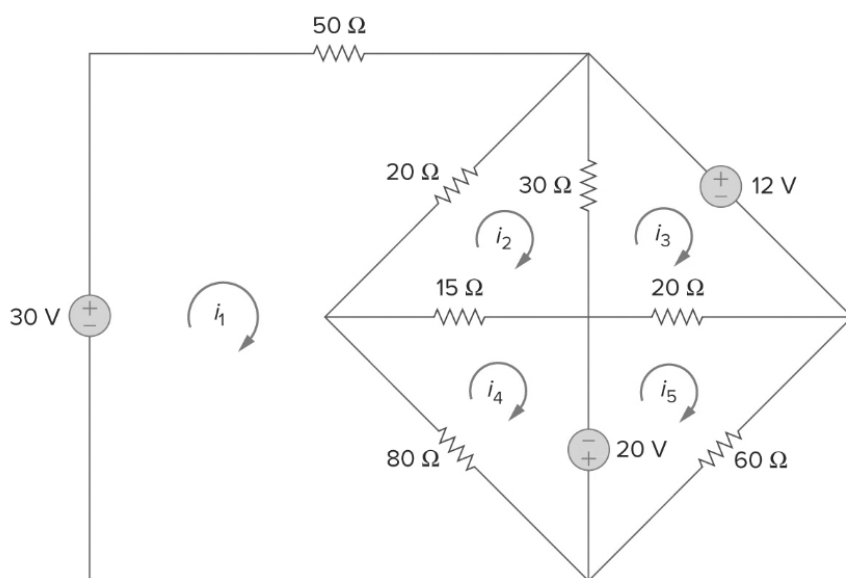
<https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html>

It is only supported on Windows and Mac. While it has been shown to work on Linux with "PlayOnLinux", that is considered "beta", i.e. unsupported.

1. Load the version appropriate to your computer - **should already have been done!**
2. Download the "LTSpice - Getting Started Guide.pdf" document
3. Perform the first 38 pages of the tutorial
  - Stop at the section titled "Generating a BOM and Efficiency Report"
  - We will cover the remainder of this guide at a later time (when we get to AC circuits)
4. Make sure you are familiar with the tool and how to create and simulate a circuit

### Part II: Simulation and Practice

The following circuit will be used for this simulation lab experiment. The figure is set up for a mesh current analysis and can be solved via the matrix methods of chapter 3. The solution involves writing and solving a  $5 \times 5$  matrix equation. Normally, I would ask you to determine the mesh equations and setup the matrices, but as this lab is more about LTSpice, I will give you the matrices to solve.



$$[I] = [R]^{-1}[V]$$

$$[R] = \begin{bmatrix} 150 & -20 & 0 & -80 & 0 \\ -20 & 65 & -30 & -15 & 0 \\ 0 & -30 & 50 & 0 & -20 \\ -80 & -15 & 0 & 95 & 0 \\ 0 & 0 & -20 & 0 & 80 \end{bmatrix}$$

$$[V] = \begin{bmatrix} 30 \\ 0 \\ -12 \\ 20 \\ -20 \end{bmatrix}$$

Note, this is just the matrix form of Ohm's law and the branch currents are given by:

- $I(50\Omega) = i_1$
- $I(20\Omega) = i_2 - i_1$
- $I(12V) = i_3$
- $I(60\Omega) = i_5$
- $I(80\Omega) = i_4 - i_1$

1. Before starting your LTSpice simulation, enter the matrices given in the above section into Scilab and compute the mesh currents and solve for the branch currents (using the above equations). These branch currents are what the LTSpice simulation will compute.
2. Start up LTSpice and create a new schematic. Save this under a filename that includes your assignment, last name, first name and the assignment, e.g. Lab03-Murthy-Purna
3. Enter the resistors, sources, and interconnects - note, all interconnects are connected (there are no “jumps”). Add ground to bottom node.
4. Change the values of the elements to correspond the values in the figure.
5. Enter the simulation command on the schematic to “DC Operating Point” (shown on your schematics as “.op”)
6. Enter a comment on the schematic that identifies this assignment and your name
7. Run the simulation - a window should pop-up that gives the voltages and currents at the nodes and in the elements respectively
8. Compare your simulations with the calculations done in step 1. Are there any differences and/or discrepancies? If so, be sure to resolve them or, alternatively, explain them in the lab report.
9. Generate a lab report with the following parts:
  - Title Page - contains lab number, lab assignment, your name, and date
  - Calculations
    - Scilab program and runtime printout
    - Equations reconciling SciLab mesh currents to the LTSpice branch currents
  - Schematic – A copy of the LTSpice schematic image you developed, including the simulation commands, comments, etc.
  - Net List - A copy of the net list (provided by LTSpice)
  - Results - A copy of the simulation run results
  - Discussion - Reconcile and/or explain any discrepancies between the calculations and simulations.

Write your report and submit a single PDF file to Blackboard. Make sure to organize your materials and make it readable.